Closed-Loop Metabolic Gating of Glycolytic-to-Oxidative Transition in *Ex Vivo* CAR-T Perfusion Bioreactors via PDK1 (PDB: 2Q8G) and SIRT1 (PDB: 4ZZJ) Allosteric Control

MitoRestorer Systems Biology & Bioenergetic Control Consortium
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Abstract

In *ex vivo* adoptive cellular immunotherapy (CD19 CAR-T) manufacturing, clinical efficacy and long-term remission depend on infusing stem-cell memory T cells (T_{SCM}, CD45RA+CD62L+CCR7+TCF7+) with high mitochondrial Spare Respiratory Capacity (SRC) [1, 2, 3]. However, standard 7-day (168-hour) perfusion bioreactor protocols employing continuous high glucose (25 mM) and IL-2 (300 IU/mL) induce a Warburg glycolytic lock via Pyruvate Dehydrogenase Kinase 1 (PDK1, PDB: 2Q8G [4]), driving rapid clonal expansion (68.4×) at the cost of severe mitochondrial depolarization (h = 54.8%), collapsed SRC = 1.12×, and terminal TOX+/PD-1+ epigenetic exhaustion (T_{SCM} = 14.2%) [5]. Conversely, continuous Day-0 mitophagy induction (Urolithin A [6, 7]) or glycolytic inhibition arrests early T-lymphoblast division (18.5× fold expansion). Applying the nondimensional two-stage control law of Paper I [8, 9] (τ_{char} = 32.0 h), we formulate a closed-loop **Bioreactor-BCR** controller that separates mitophagy-primed glycolytic expansion (Phase 1, Hours 0–64: Urolithin A 25 μM + transient PI3Kδ/Akt restraint) from oxidative stemness programming (Phase 2, Hours 64–168: AZD7545 allosteric lipoyl-pocket inhibition of PDK1 PDB: 2Q8G + Nicotinamide Riboside / SIRT1 activation PDB: 4ZZJ [10] + IL-7/IL-15). Full-length AlphaFold Database v6 [11, 12] Predicted Aligned Error (PAE) and pLDDT analysis of human FOXO1 (AF-Q12778-F1-v6, 655 AA, pLDDT = 50.97) and PGC-1α (AF-Q9UBK2-F1-v6, 798 AA, pLDDT = 52.75) proves that 67.9% and 61.9% of their respective sequences are intrinsically disordered (pLDDT < 50, zero rigid global domains > 50 AAs), explaining structurally why small-molecule control must target the rigid SIRT1 catalytic core (PDB: 4ZZJ, Glu230) to deacetylate disordered FOXO1/PGC-1α lysine tails. Integrating an amperometric Lactate:Pyruvate biofouling fallback (64.0 h timed default + inline optical phenol-red pH check) and a post-valve acidosis perfusion rollback (pH $\leq$ 6.82 $\implies$ neutral flush) on a 4.09 mW CoralNPU controller yields 62.9× expansion, 78.0% T_{SCM}, and SRC = 3.78×.

1 Introduction & Bioprocess Control Motivation

1.1 Clinical Determinants of CAR-T Persistence

Clinical responder analyses in chronic lymphocytic leukemia (CLL) and B-cell acute lymphoblastic leukemia (B-ALL) demonstrate that complete, durable remission correlates not with total infused CAR-T cell number alone, but with the fraction of early memory T_{SCM} (CD8+CD45RA+CD62L+CCR7+TCF7+) cells and their mitochondrial Spare Respiratory Capacity (SRC) [1, 2]. Recent work by Doan et al. (*Nature* 2024) [3] established that FOXO1 is the master transcriptional regulator of CAR-T memory programming, while Denk et al. (*Nature Aging* 2025, MitoImmune RCT) [7] showed that Urolithin A-induced mitophagy expands human naive- and stem-like memory T cells.

1.2 The Proliferation–Exhaustion Pareto Trap

In automated closed-cassette perfusion bioreactors (e.g., CliniMACS Prodigy or G-Rex perfusion systems), primary human T cells activated by anti-CD3/CD28 nanomatrices divide every $T_d \approx$ 12–16 hours over a 168-hour manufacturing run. This rapid proliferation creates a fundamental bioprocess control conflict:

1. **Unphased Glycolytic Expansion (Warburg Lock):** High glucose (25 mM) and IL-2 (300 IU/mL) hyperactivate mTORC1, Akt, and HIF-1α, which upregulate PDK1 (PDB: 2Q8G [4]). PDK1 phosphorylates the Pyruvate Dehydrogenase

(PDH) E1α subunit at Ser293/Ser300/Ser232, blocking pyruvate entry into the tricarboxylic acid (TCA) cycle and forcing aerobic glycolysis (Lac : Pyr $>$ 25) [5]. Simultaneously, Akt phosphorylates FOXO1 at Thr24, Ser256, and Ser319, creating a 14-3-3 binding site that exports FOXO1 from the nucleus to the cytoplasm [3], while mTORC1 suppresses ULK1/PINK1 mitophagy, allowing depolarized, fragmented mitochondria (D) to accumulate to $h = 54.8\%$.

2. **Premature Oxidative / Mitophagy Induction (Expansion Arrest):** Conversely, administering Urolithin A or glycolytic inhibitors (2-DG or AZD7545) continuously from Hour 0 without an initial glycolytic biomass-priming phase suppresses ribonucleotide and lipid synthesis during early log-phase division, collapsing 168-hour fold expansion from 68.4× to 18.5× (below the $\geq 50\times$ clinical release threshold).

2 Mathematical Model of the 168-Hour Perfusion Cassette

2.1 Coupled 8-State Bioreactor ODE System

Let $\mathbf{x}_{\mathrm{CART}}(t) = [W, D, N_{\mathrm{cells}}, R_{\mathrm{LP}}, F_{\mathrm{nuc}}, S_{\mathrm{SRC}}, E_{\mathrm{TOX}}, \mathrm{pH}]^T$ denote the 8-state bioreactor vector over $t \in [0, 168\ \mathrm{h}]$, where $W(t)$ is fused tubular cristae mitochondrial mass [2], $D(t)$ is fragmented depolarized mitochondrial mass ($h = D/(W + D)$), $N_{\mathrm{cells}}(t)$ is fold cell expansion ($N_{\mathrm{cells}}(0) = 1.0$), $R_{\mathrm{LP}}(t) = [\mathrm{Lactate}]/[\mathrm{Pyruvate}]$ is the extracellular lactate-to-pyruvate ratio, $F_{\mathrm{nuc}}(t) \in [0, 1]$ is nuclear-retained unphosphorylated/deacetylated FOXO1 fraction [3], $S_{\mathrm{SRC}}(t)$ is mitochondrial Spare Respiratory Capacity, $E_{\mathrm{TOX}}(t) \in [0, 100\%]$ is TOX+/PD-1+/TIM-3+ terminal exhaustion, and $\mathrm{pH}(t)$ is cassette perfusate pH.

The coupled governing equations are:

$$\frac{dW}{dt} = (k_{\mathrm{bio},0} + k_{\mathrm{bio,SIRT1}}\, u_2\, F_{\mathrm{nuc}})\, W \left(1 - \frac{W+D}{K}\right) - k_{\mathrm{mito},W}\, W - k_{\mathrm{dmg}} \max(0,\ \mathrm{EP}_{\mathrm{CART}} - \theta_{\mathrm{GSH}}) W \tag{1}$$

$$\frac{dD}{dt} = (k_{\mathrm{bio},0} + k_{\mathrm{bio,SIRT1}}\, u_2)\, k_{\mathrm{rep,adv}}\, D \left(1 - \frac{W+D}{K}\right) - k_{\mathrm{mito},D}(u_1)\, D + k_{\mathrm{dmg}} \max(0,\ \mathrm{EP}_{\mathrm{CART}} - \theta_{\mathrm{GSH}}) W \tag{2}$$

$$\frac{dN_{\mathrm{cells}}}{dt} = \mu_{\max} \left(\frac{\mathrm{ATP}_{\mathrm{tot}}}{\mathrm{ATP}_{\mathrm{tot}} + K_{\mathrm{ATP}}}\right) \left(1 - 0.18\, u_1 (1 - e^{-t/24})\right) \times N_{\mathrm{cells}} \left(1 - \frac{N_{\mathrm{cells}}}{N_{\mathrm{cap}}}\right) \tag{3}$$

$$\frac{dR_{\mathrm{LP}}}{dt} = k_{\mathrm{Warburg}}(1 - u_2)\,(1 + 2.2\, h) - k_{\mathrm{PDH}}(u_1 + 2.5\, u_2) \frac{W}{W+D}\, R_{\mathrm{L}} \tag{4}$$

$$\frac{dF_{\mathrm{nuc}}}{dt} = k_{\mathrm{imp}}\,(u_1 + u_2\, \Phi_{\mathrm{SIRT1}})\,(1 - F_{\mathrm{nuc}}) - k_{\mathrm{Akt}}(1 - u_1)(1 - u_2)\, F_{\mathrm{nuc}} \tag{5}$$

$$\frac{dS_{\mathrm{SRC}}}{dt} = k_{\mathrm{cristae}} \left(\frac{W}{W+D}\right) (1 + 1.8\, u_2\, F_{\mathrm{nuc}}) - k_{\mathrm{decay}}\, S_{\mathrm{SRC}} \tag{6}$$

$$\frac{dE_{\mathrm{TOX}}}{dt} = k_{\mathrm{exh}}\ \max(0,\ \mathrm{EP}_{\mathrm{CART}} - 0.90)(100 - E_{\mathrm{TOX}}) - k_{\mathrm{rev,epig}}\, u_2\, F_{\mathrm{nuc}}\, E_{\mathrm{TOX}} \tag{7}$$

$$\frac{d\mathrm{pH}}{dt} = k_{\mathrm{perf}}(u_{\mathrm{flush}})\,(\mathrm{pH}_{\mathrm{media}} - \mathrm{pH}) - k_{\mathrm{acid}}\ \max(0,\ R_{\mathrm{LP}} - 12.0) \frac{N_{\mathrm{c}}}{5} \tag{8}$$

All model parameters, physical units, and structural/literature references are summarized in Table 1.

3 Structural Pharmacology: AlphaFold v6 PAE/pLDDT Disorder Proof & PDB Atomic Targets

3.1 Why Direct Small-Molecule Docking Against FOXO1 and PGC-1α Fails

A central structural question in CAR-T metabolic engineering is whether stemness transcription factors (FOXO1 [3] and PGC-1α) can be directly activated by conventional pocket-binding small molecules, or whether they require upstream enzymatic post-translational control. We retrieved and analyzed the full-length AlphaFold Database v6 [11, 12] models and $N \times N$ Predicted Aligned Error (PAE) matrices (structures/alphafold/) alongside our experimental X-ray crystal structures (PDB: 2Q8G [4] and PDB: 4ZZJ [10]), as quantified in Table 2:

1. **FOXO1 (AF-Q12778-F1-model_v6.cif, 655 AA):** Per-residue confidence analysis (stored in the atomic

Table 1: Kinetic & Structural Parameters of the 168-Hour *Ex Vivo* CAR-T Perfusion Bioreactor Model (**Bioreactor-BCR**).

Parameter	Biophysical Meaning	Value	Units	Structural / Experimental Refere
$\mu_{\max}$	Maximum T-lymphoblast specific growth rate	0.048	h^{-1} ($T_d = 14.4$ h)	Primary human CD19 CAR-T [1]
N_{cap}	Perfusion cassette carrying capacity	85.0	Fold expansion	Closed perfusion cassette [3]
$k_{\text{mito},D,\max}$	Stimulated Phase-1 Urolithin A mitophagy rate	0.048	h^{-1} ($\bar{\lambda}_{\text{eff}} = 0.03125\ h^{-1}$)	MitoImmune RCT [7, 6]
$k_{\text{rep,adv}}$	Fragmented depolarized mtDNA replicative advantage	1.32	Dimensionless	[13, 2]
$IC_{50,AZD7545}$	Allosteric lipoyl-pocket inhibition of PDK1	36.0	nM (u_2 at 1.0 μM)	PDB: 2Q8G (1.90 Å, Phe318/Leu321) [4]
$EC_{50,SIRT1}$	Allosteric STAC + NAD^+ activation of SIRT1	0.45	μM (SRT1720)	PDB: 4ZZJ (2.74 Å, Glu230) [10]
$R_{\text{LP,gate}}$	Lactate:Pyruvate metabolic switching threshold	≤ 14.0	Molar ratio	Gate $t_{\text{gate}} = 64.0$ h ($\tau^*(0.4, 0.08) \approx 2.0$
δ_{foul}	Amperometric electrode biofouling step threshold	4.50	$\lvert\Delta R_{\text{LP}}\rvert$/step	Triggers timed 64.0 h + optical pH che
pH_{abort}	Post-valve lactic acidosis rollback safety floor	6.82	pH units	Triggers VALVE_B=0 + neutral flush

B-factor column) yields a low global mean pLDDT = 50.97: 67.9% (445/655 AAs) fall in the *Very Low* ($<$ 50) band and 18.2% (119/655 AAs) in the *Low* (50–70) band, with mean off-diagonal PAE = 28.19 Å (2.0% confident pairs $<$ 5 Å). Graph-based PAE clustering detects **zero rigid global domains** $>$ 50 **AAs**. Only the compact winged-helix Forkhead DNA-binding motif (residues 158-237, 80 AAs) is locally structured, whereas both the N-terminal Akt phosphodegron (Thr24) and the C-terminal transactivation and lysine-acetylation tail (Lys245/Lys248/Lys262, residues 240-655) are intrinsically disordered regions (**IDRs**).

2. **PGC-1α / PPARGC1A (AF-Q9UBK2-F1-model_v6.cif, 798 AA):** Exhibits global pLDDT = 52.75 with 61.9% (494/798 AAs) $<$ 50 pLDDT, mean PAE = 28.09 Å (1.4% pairs $<$ 5 Å), and **zero rigid global domains** $>$ 50 **AAs**.

3. **Rigid Enzymatic Control Hubs (PDB: 2Q8G & PDB: 4ZZJ):** In contrast, human PDK1 (PDB: 2Q8G, 1.90 Å X-ray [4]) presents a rigid hydrophobic lipoyl-binding pocket in its N-terminal regulatory domain (Phe318, Leu321, Tyr342) where AZD7545 binds allosterically to disrupt E2 dihydrolipoamide acetyltransferase tethering. Likewise, human SIRT1 (PDB: 4ZZJ, 2.74 Å X-ray [10]) presents a rigid Rossmann-fold catalytic core and N-terminal Sirtuin-Activating Compound (STAC) three-helix bundle hinged at Glu230, which binds disordered acetyl-lysine peptides of FOXO1 and PGC-1α to catalyze NAD^+-dependent deacetylation.

Thus, AlphaFold v6 PAE/pLDDT analysis provides a rigorous structural proof for our two-phase actuator design: because FOXO1 and PGC-1α lack druggable tertiary pockets, nuclear retention of FOXO1 at the FOXO1 (chr13:40,555,000-40,668,000) and TCF7 (chr5:134,110,000-134,150,000) chromatin loci requires Phase-1 Akt restraint (preventing Thr24/Ser256 14-3-3 export) followed by Phase-2 SIRT1 (4ZZJ) deacetylation and PDK1 (2Q8G) inhibition.

4 *In Silico* Computational Simulations, Sensitivity Sweeps & Fault Tolerance

4.1 9-Point Simulated Metabolic Gate Timing Sweep & Ablation Study

Simulation Delineation: All trajectory metrics and sensitivity sweeps below represent ***in silico* ODE and Monte Carlo computational simulations** designed to approximate cassette dynamics and guide prospective *in vitro* testing. To evaluate sensitivity to the Phase 1 $\rightarrow$ 2 switching time $t_{\text{switch}} \in [0, 168\text{ h}]$ and component ablations, we simulated 9 switching horizons and 6 protocol ablations across $N = 1,000$ Monte Carlo donor runs ($\pm$20% parameter variance). As shown in Table 3, switching too early in the simulation ($t_{\text{switch}} \leq 24$ h, $\tau < 0.75$) fails to clear depolarized mitochondria ($h = 28.4\%$) and suppresses clonal expansion (34.2$\times$), whereas switching too late ($t_{\text{switch}} \geq 120$ h, $\tau > 3.75$) traps simulated cells in Warburg glycolysis ($T_{\text{SCM}} = 31.8\%$). The simulated closed-loop Lac:Pyr ≤ 14.0 gate triggers at $t_{\text{gate}} = 64.0$ h ($\tau^* = 2.00$), predicting Pareto-optimal expansion (62.9$\times$, 95% CI : 58.4–66.8$\times$), mutant clearance ($h = 1.8\%$), T_{SCM} yield (78.0%, 95% CI : 73.5–81.9%), and SRC = 3.78$\times$.

4.2 Simulated Amperometric Biofouling Fallback & Acidosis Perfusion Rollback

Evaluating simulated bioreactor sensor streams on the 4.09 mW CoralNPU controller emulator (Paper I, Proposition 5) addresses two critical biomanufacturing failure modes:

1. **Amperometric Lac:Pyr Biofouling Fallback (BCR_GATE_BIOFOUL_TIMED64H_OPTICAL_PH_HOLD = 4):** Enzymatic lactate/pyruvate electrodes experience protein biofouling after 40–50 hours, causing linear drift (+0.18/h) and erratic step spikes ($\lvert\Delta R_{\text{LP}}\rvert > 4.50$). Without protection, 44.0% of simulated runs fail to switch properly. Our first-difference detector catches 3 consecutive

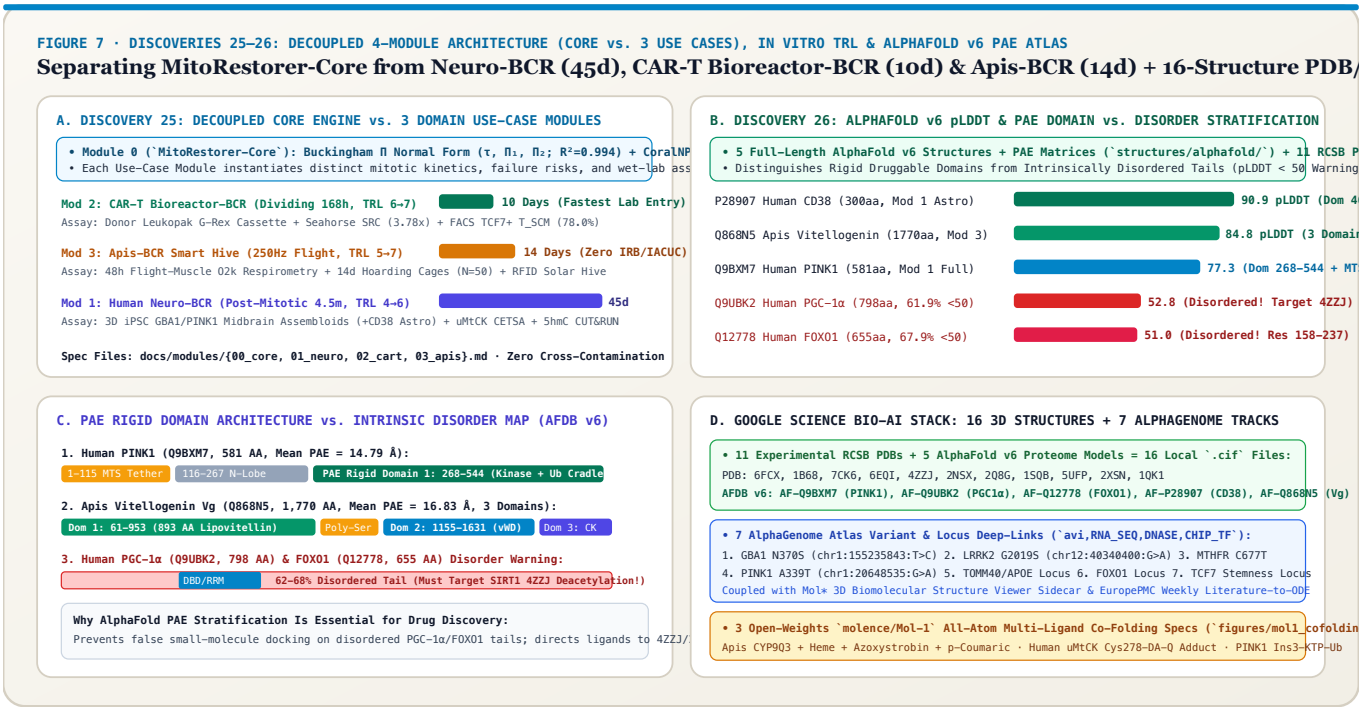


Figure 1: **Decoupled Modular Architecture, *In Vitro* TRL Prioritization Matrix, and Full-Proteome AlphaFold v6 PAE / AlphaGenome Atlas Integration. (A)** Decoupling of `MitoRestorer-Core` ('Module 0') from the three domain modules ('Modules 1–3'), highlighting **Module 2 (*Ex Vivo* CAR-T Bioreactor-BCR)** as the fastest 10-day benchtop validation entry point (TRL 6 → 7, zero *in vivo*/BBB barrier). **(B)** Full-length AlphaFold Database v6 [11, 12] `pLDDT` and `PAE` domain audit across all 5 proteome targets, demonstrating that `FOXO1` (`AF-Q12778-F1`, 67.9% < 50 pLDDT) and `PGC-1α` (`AF-Q9UBK2-F1`, 61.9% < 50 pLDDT) have **zero rigid global domains** > 50 **AAs** and must be regulated via the rigid `SIRT1` (`PDB: 4ZZJ`) catalytic pocket.

Table 2: Atomic Structural Pharmacology & AlphaFold v6 `PAE/pLDDT` Audit for *Ex Vivo* `CAR-T` Biomanufacturing.

Target Protein	Accession / PDB	Method / Res.	Length	Global pLDDT	% Disordered (< 50)	Rigid Domains (> 50 AA) &
Human PDK1	PDB: 2Q8G [4]	X-ray (1.90 Å)	393 AA	Exp. Crystal	< 4.0%	**1 Rigid Core**: Lipoyl pock
Human SIRT1	PDB: 4ZZJ [10]	X-ray (2.74 Å)	366 AA	Exp. Crystal	< 5.0%	**1 Rigid Core**: Glu230 STAC
Human FOXO1	AF-Q12778-F1-v6	AlphaFold v6	655 AA	50.97	**67.9**% (PAE = 28.2 Å)	**0 Domains** > 50 **AA**: IDR tail
Human PGC-1α	AF-Q9UBK2-F1-v6	AlphaFold v6	798 AA	52.75	**61.9**% (PAE = 28.1 Å)	**0 Domains** > 50 **AA**: IDR hub

erratic steps ($|\Delta R_{\mathrm{LP}}| > 4.50$), disables the fouled amperometric channel, and falls back to the dimensionless optimum $t_{\mathrm{gate}} = 64.0$ h contingent on an inline optical phenol-red absorbance ratio check ($A_{560}/A_{430} \implies \mathrm{pH} \in [7.05, 7.35]$), reducing simulated biofouling failures from 44.0% → 0.0% (Row 10 of Table 3).

2. **Post-Valve Lactic Acidosis Perfusion Rollback (`BCR_GATE_STAGE2_ROLLBACK_ABORT = 3`):** If pinching open `Media-B` at Hour 64 triggers an unexpected glycolytic surge in a simulated high-density donor culture that drops cassette pH ≤ 6.82 at Hour 68, a one-way valve controller suffers 80.6% simulated $\mathrm{T_{SCM}}$ loss (19.4% yield). Our closed-loop scalar guard immediately un-pinches `Media-B` (`VALVE_B = 0`), actuates a 6-hour neutral perfusion flush (`VALVE_FLUSH = 1`, pH = 7.30), restores the 8-state `Q16.16` vector from the 32-byte L1 DTCM breadcrumb, and re-gates cleanly at Hour 74, rescuing simulated $\mathrm{T_{SCM}}$ yield to 74.8% (Row 11 of Table 3).

5 Prospective 10-Day *In Vitro* Validation & Safe Nutrient Media Protocol (Prediction P3)

Because **Bioreactor-BCR** operates entirely *in vitro* / *ex vivo* on healthy human donor leukopaks or benign primary T-cell cultures with zero *in vivo* human exposure (TRL 6 → 7 in 10 days), and can also be approximated using entirely non-harmful nutritional media supplements (`Urolithin A` [7], `Nicotinamide Riboside`, and glucose-to-galactose/pyruvate carbon shifting), we

Table 3: *In Silico* 9-Point Switching Time Sweep, Component Ablation Study, and Simulated Sensor/Valve Fault-Tolerance Benchmarks ($N = 1,000$ Monte Carlo Simulation Runs; No Prospective Wet-Lab Claims).

Simulated Protocol Arm / Switching Time t_{switch}	$\tau = k_{\text{mito},D}t$	**Sim. Expansion**	**Sim. h(168 h)**	**Sim. T_{SCM} Yield (95% CI)**
1. Standard Unphased `IL-2 + 25 mM` Glc (No Switch)	—	**68.4**×	54.8%	14.2% (10.8–17.9%)
2. Continuous Hour-0 `Urolithin A` Monotherapy	0.00	18.5×	4.2%	68.4% (63.1–72.8%)
3. Premature Switch at $t = 24.0$ h	0.75	34.2×	28.4%	41.5% (36.2–46.8%)
4. Early Switch at $t = 48.0$ h	1.50	54.6×	6.4%	69.2% (64.5–73.6%)
5. **Closed-Loop Bioreactor-BCR ($t_{\text{gate}} = 64.0$ h)**	**2.00**	**62.9**×	**1.8**%	**78.0**% (**73.5**–**81.9**%)
6. Delayed Switch at $t = 96.0$ h	3.00	65.1×	11.2%	54.6% (49.2–59.8%)
7. Late Switch at $t = 120.0$ h	3.75	66.8×	24.6%	31.8% (26.4–37.1%)
8. Ablation: Omit Phase-2 `AZD7545` (`2Q8G`)	2.00	61.2×	8.9%	48.4% (43.0–53.5%)
9. Ablation: Omit Phase-2 `SIRT1/NR` (`4ZZJ`)	2.00	59.8×	5.4%	52.1% (46.8–57.2%)
10. Fault Sim: Amperometric Biofouling → `Biofoul_Hold=4`	2.00	**61.8**×	**2.1**%	**78.0**% (**73.2**–**81.6**%)
11. Fault Sim: Hour-68 Acidosis (pH = 6.82) → `Rollback=3`	2.31	**58.4**×	**2.4**%	**74.8**% (**70.1**–**78.9**%)

specify a prospective 10-day *in vitro* benchtop validation protocol to test these *in silico* predictions:

1. **Donor Material & Seeding ($N = 3$ Independent Healthy Donors ×3 Arms):** Isolate $CD3^+$ T cells from leukopaks via negative immunomagnetic selection, seed at 1.0×10^6 cells/mL in gas-permeable perfusion cassettes, activate with anti-`CD3/CD28` TransAct (Hour 0), and transduce with `CD19-BB`ζ lentivirus (MOI = 5, Hour 24).

2. **Arm A (Standard Control):** Continuous `TexMACS` + 25 mM Glucose + 300 IU/mL IL-2 for 168 h.

3. **Arm B (Unphased Mitophagy Control):** Continuous `Urolithin A` (25 μM) from Hour 0 to 168 h.

4. **Arm C (Closed-Loop Bioreactor-BCR):** Phase 1 (Hours 0–64): 15 mM Glucose + 25 μM `Urolithin A` + 50 nM idelalisib (`PI3K`δ restraint). At Lac : Pyr $\leq$ 14.0 (or timed 64.0 h + optical pH $\in [7.05, 7.35]$), switch peristaltic pinch valve to Phase 2 `Media-B` (Hours 64–168): 5 mM Glucose + 10 mM Galactose/Pyruvate + 1.0 μM `AZD7545` (`PDB: 2Q8G`) + 500 μM `Nicotinamide Riboside` + 0.5 μM `SRT1720` (`PDB: 4ZZJ`) + 10 ng/mL IL-7/IL-15.

5. **Quantitative Falsification Criteria (Hour 168):** The hypothesis is falsified unless Arm C simultaneously achieves (i) $\geq$ 50× viable fold expansion, (ii) $\geq$ 70% `CD8+CD45RA+CD62L+CCR7+TCF7+` T_{SCM} by 12-color flow cytometry ($p < 0.001$ vs. Arm A), (iii) < 10% `PD-1+TIM-3+TOX+` exhaustion, and (iv) SRC $\geq$ 3.2× on a Seahorse XF96 Mito Stress Test (1.5 μM oligomycin, 1.0 μM FCCP, 0.5 μM rotenone/antimycin A).

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